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Feed-Through Intervention Window Adds Versatility on Intelligent Completion Multilateral Wells

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Abstract

For three decades multilateral well construction has been widely adopted in the North Sea and the Middle East as a beneficial method of field development. Further opportunities are available to increase production performance throughout the life of multilateral wells with the implementation of variable inflow control and monitoring for the mainbore and laterals while maintaining lateral access.

An operator in the Middle East, with experience in both multilateral and intelligent completion system installations, wanted to combine the benefits of both completion strategies to further enhance well economics and extend field production life. The solution supports several intelligent completion options through integrating control line feed-through capability into a field-proven through-completion intervention multilateral window system. This paper highlights the development, testing, and a successful case study of the integrated system for an intelligent multilateral application in the Middle East.

The paper discusses how through-completion intervention access to the multilateral branches was achieved, while adding independent flow control and permanent lateral monitoring. The paper emphasizes how wells originally planned as single wellbores, or simple dual laterals can now benefit from intelligent flow control in both the mainbore and lateral zones. The methodology for continually improving the reliability and versatility of multilateral installations to maximize efficiencies is also highlighted. The case study illustrates the practical implementation and success of intelligent multilateral strategies in the region. The benefits of combining reservoir control with multilateral technology are highlighted, including reduced CAPEX and OPEX with a lower carbon footprint. The paper emphasizes how intelligent multilateral field strategies can optimize hydrocarbon access and recovery while minimizing construction hours and costs.

The case study discusses the first installation in the region where an operator has taken advantage of a field-proven through-completion intervention multilateral window system integrated with control line feed-through capability to facilitate control and monitoring of each reservoir section. The paper provides a comprehensive overview of the advancements, applications, and benefits of intelligent multilateral technology in the Middle East, with a focus on reducing costs, and optimizing asset value.

Introduction

Multilateral installations have demonstrated sustained success in the Middle East for over two decades. The region saw its first multilateral well installed in the early 1990s, with local operators quickly emerging as pioneers in adopting and validating multilateral development as a reliable strategy to enhance hydrocarbon recovery. These wells improve reservoir exposure by enabling access to multiple production targets through a single main wellbore.

The earliest multilateral wells were basic in design, typically consisting of dual laterals with simple completions. These configurations offered limited or no intervention capabilities and performing workover operations on such wells required full Decompletion. Laterals were not accessible with tubing after they were drilled and served only to provide a path for drainage in additional productive sections (DeMong et al., 2000)

Over time, multilateral systems have continued to evolve to address the growing challenges faced by operators. Ongoing innovation and refinement have expanded system capabilities, offering a range of completion solutions that prioritize both productivity and operational efficiency. Today's multilateral wells can be constructed as new installations or retrofitted from existing single-bore wells. Modern systems now offer full pressure isolation at the junction, along with integrated monitoring and flow control capabilities.

The latest generation of multilateral technologies includes features such as segregated through-tubing access to each lateral, advanced flow control, pressure-isolated junctions, and full workover capability—enabling operators to upgrade completions at any point in the well's life cycle (Kelsey et al., 2022)

Current innovations continue to push boundaries, focusing on the integration of advanced completion technologies, streamlined operations, reduced system complexity, and improved field development viability. As a result, Multilateral wells, Intelligent Completions, and Inflow Control Devices have gained prominence as effective solutions for increasing ultimate recovery while lowering the cost per barrel (Shahreya et al., 2021)

Case Study

This case study focuses on an operator in the Middle East with experience deploying Multilateral Intervention Window Systems (Ameen et al., 2023), both in dual-lateral producer wells and in configurations where the lateral serves as the primary producer and while the mainbore is utilized for reservoir monitoring. This multilateral completion system incorporates both unshrouded and shrouded interval control valves (ICVs) installed above the junction, allowing for independent flow control of the mainbore and lateral branches. This approach has proven effective in providing selective zonal management and enhanced reservoir access.

Challenges

The operator was faced with the challenge of integrating previously independent capabilities into a cohesive multilateral completion solution. The objective was to use the mainbore as an observation well and the lateral for production. Key requirements included rigless intervention access for future remedial operations and the ability to implement near-reservoir flow control and monitoring. These criteria highlighted the need for a reliable method of running control lines across a multilateral window.

To address this, the operator pursued an evolved solution by incorporating control line feed-through capability into the long-standing, field-proven through-completion intervention multilateral window system, which has been successfully deployed in the Middle East for over 25 years. The upgraded system maintains the original components used for lateral access and control, while adding the capability to interface with intelligent completion tools. This enables the deployment of flow control valves and permanent

monitoring systems below the window, closer to the reservoir, supporting enhanced production management and data acquisition.

To support this strategy, the following operational and system challenges have been outlined:

Independent Flow Control and Monitoring

Independent flow control and monitoring for oil wells offers several key benefits enhancing operational efficiency, safety and resource management. Precise regulation of flow rates from individual wellbores or zones maximizes hydrocarbon recovery while minimizing water and/or gas breakthrough. This tailored approach enhances overall production efficiency by leveraging real-time monitoring to deliver valuable insights into the reservoir's health and behavior, enabling more informed and timely decision making. This helps operators adjust production strategies to extend the life of the reservoir and improve recovery rates.

Continuous monitoring detects anomalies like pressure spikes or equipment failures early, reducing well control risks or other hazardous incidents. Automated flow control systems minimize the need for manual interventions, lowering labor costs and downtime. By controlling flow to prevent excessive gas or water production, these systems reduce waste and their environmental footprint of operations - including minimizing flaring or surface storage and processing. Advanced sensors and data analytics provide actionable insights enabling operators to make informed decisions quickly such as adjusting choke settings or identifying maintenance needs.

In wells with multiple zones of interest, independent monitoring and flow control allows operators to manage each zone separately, optimizing output from high performing zones while addressing issues in others. Accurate monitoring ensures compliance with regulatory requirements by providing reliable data for production reporting and environmental monitoring. Positioning flow control and reservoir monitoring equipment near the mainbore reservoir helps ensure accurate data acquisition. These benefits collectively lead to more efficient, safer, and sustainable operations with improved economic and environmental outcomes.

Rigless Lateral Access

Rigless lateral access in multilateral wells refers to the ability to access and manage lateral branches without requiring a drilling rig, typically using wireline, coiled tubing, or other intervention technologies. This approach offers significant benefits, particularly in terms of cost, efficiency, and operational flexibility.

Eliminating the need for a drilling rig significantly reduces operational costs, including rig mobilization, day rates, and associated logistics. Rigless interventions are less resource intensive lowering overall expenses. Rigless access also allows for quicker interventions, such as re-entering laterals for maintenance, stimulation, or data collection. This minimizes downtime and keeps production online, maximizing the well productivity.

Operators can selectively access specific laterals to perform targeted operations like cleaning out blockages or performing workovers without affecting other laterals. Rigless interventions enable rapid response to problems like water breakthrough, sand production, or equipment failures in specific laterals, improving well performance and reducing risk of production losses. This adaptability optimizes production from each lateral.

By avoiding rig deployment, rigless access reduces surface footprint and environmental impact making it ideal for offshore platforms with limited space or environmentally sensitive areas. Fewer personnel and less heavy equipment are required compared to rig-based operations, lowering the risk of accidents and improving overall safety on the well site.

In multilateral wells, rigless access provides a practical solution for managing complex architectures allowing operators to maintain and optimize productions across all laterals efficiently. Rigless lateral access in multilateral wells offers cost-effective, flexible, and environmentally friendly solutions for managing

complex well systems, improving production efficiency and extending well life, while minimizing operational risk and disruptions.

Solution

The integration of control line feed-through pathways for control lines and thermoelectric cables (TEC) into the field-proven Multilateral Intervention Window (Figure 1) enhances its suitability for complex multilateral completions. This enhanced configuration enables direct interfacing with intelligent completion tools, allowing flow control valves and permanent downhole monitoring systems (PDHMS) to be placed below the junction, closer to the reservoir (Kelsey et al., 2024)



Figure 1—Through-Completion Intervention Feed-Thru Multilateral Window System

Another key innovation is a self-alignment feature. This is particularly critical in intelligent completion environments, where downhole tool rotation in the presence of control lines and thermoelectric cables (TEC) poses a significant risk of damage to the completion string. The self-alignment feature enables up to 180 degrees of rotation in either direction, effectively eliminating the risk of mechanical interference or damage during installation. The combination of orientation and anchoring in a single motion ensures accurate space-out of completion components and operational efficiency.

The first field deployment of the integrated feed-through system was successfully completed in the Middle East in 2023. The system was installed equipped with Interval Control Valves (ICVs) (Figure 2) and Permanent Downhole Monitoring Systems (PDHMS) installed above and below the junction to support independent production control and zonal surveillance.



Figure 2—Interval Control Valves (ICV)

To manage wellbore isolation, a fluid loss control device was incorporated into the system. This device can be shifted open to enable controlled access into the mainbore lower completion section, further supporting selective production and future intervention operations.

The lower ICV and PDHMS were positioned below the junction in the mainbore, enabling near-reservoir monitoring and flow control, while the upper ICV and PDHMS were installed above the junction, maintaining full system controllability across both well branches.

Installation

The following method was used to create and install the multilateral intervention junction:

The casing string—incorporating a latch coupling and mule shoe sleeve for depth and orientation control—was run in hole and cemented in place. Using the latch coupling as an anchor and depth reference, a track-guided mill was deployed to create a near-rectangular window at the desired depth and orientation ([Kelsey et al., 2022](#)) After milling, the assembly was pulled out of hole.

Next, a milling whipstock with bolted mills was run in hole and latched into the anchor. The bolted mills were sheared from the whipstock and used to drill the initial section of the lateral. A separate drilling assembly was then deployed to extend the lateral to total depth (TD).

A liner was then run into the lateral and cemented to isolate the junction area. The second section was then drilled and completed.

Finally, the extended upper portion of the lateral liner—protruding into the mainbore—was cut flush with the wellbore and retrieved, along with the milling whipstock, using a washpipe barrel.

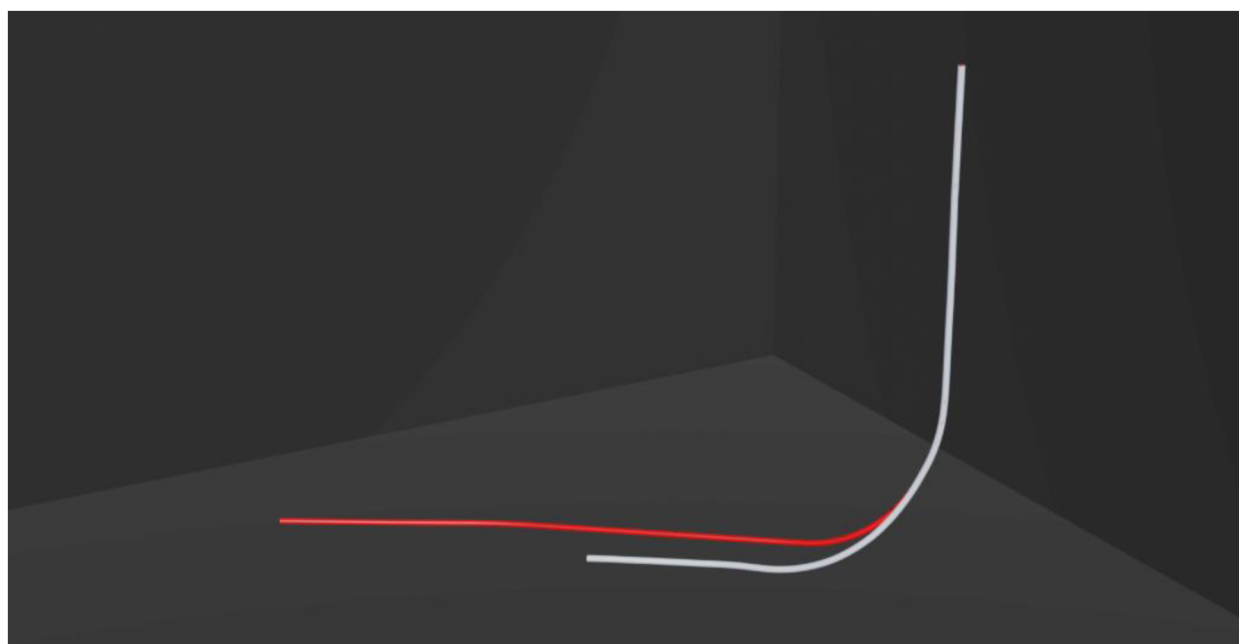


Figure 3—Wellpath

Completion

The one-trip intelligent completion system incorporating a lateral access window was successfully deployed, following a structured sequence of operations:

Pre-Completion Activities

A wellbore clean-out run was conducted to remove residual debris and ensure the wellbore met completion readiness criteria. All completion components were drifted in advance to confirm internal clearances and prevent potential deployment issues.

Completion Installation

The intelligent completion system, pre-assembled with the lateral access window, was run in hole with the production tubing. Deployment was temporarily halted 30 feet above the planned setting depth to initiate orientation procedures. As the completion string is lowered, the orienting key was aligned with the mule shoe sleeve, allowing the latching key to engage the latch coupling profile. Set-down weight was applied to confirm positive latching. A control line integrity test was then performed to validate control lines and TEC connectivity. The latch assembly was released from the mule shoe by applying controlled overpull.

Subsequently, the tubing hanger, space-out pup joints, and the Tubing Retrievable Safety Valve (TRSV) were made up and deployed. PDHMS cable readings were verified and control line continuity reconfirmed. The tubing hanger was successfully landed in the wellhead.

Lateral Access Operations

Prior to setting the production packers, lateral access operations were successfully executed. A slickline-deployed retrieving tool was used to engage and extract the existing isolation sleeve installed in the lateral access window through controlled overpull. A tubing exit whipstock was then run on slickline and accurately set at the designated profile to enable lateral re-entry. A drift run was performed 500 feet into the lateral as verification method to confirm unobstructed access.

Following this, the tubing exit whipstock was retrieved to surface using a slickline retrieval tool. Finally, a new isolation sleeve was installed via slickline and successfully set within the lateral access window, thereby restoring the mainbore pressure isolation integrity.

Final Completion Steps

Production packers were set by applying pressure against the closed isolation valve positioned below the lower packer. The Blowout Preventer (BOP) was removed, and the Christmas tree was installed. Intelligent Completion Valves (ICVs) were function tested and set to their operational positions in accordance with the approved well program.

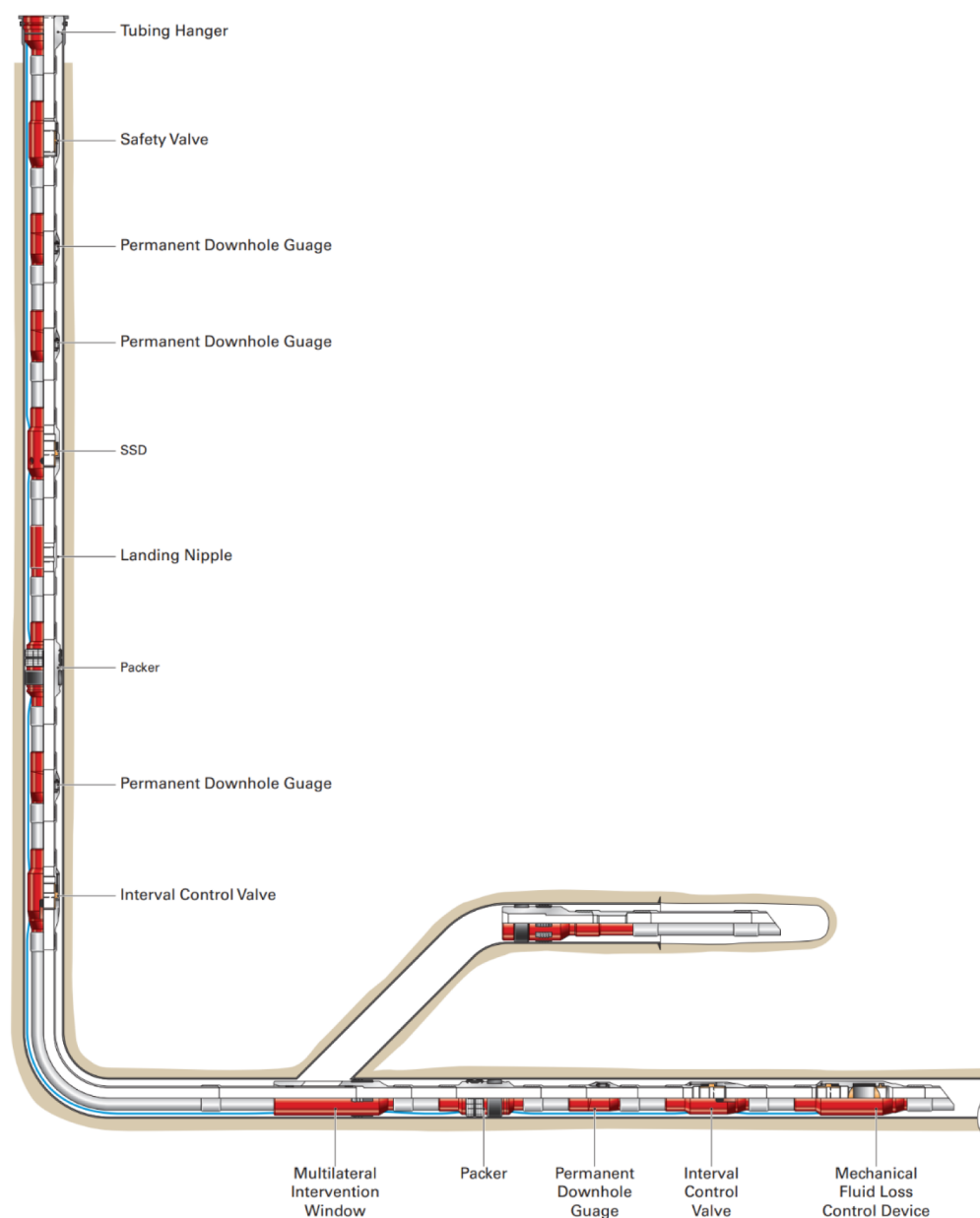


Figure 4—Through-Completion Intervention Feed-Thru Multilateral Window System Installation Schematic

This deployment not only confirmed the reliability of the integrated system but also established a new industry benchmark for intelligent multilateral completions in complex reservoir environments. The successful development and installation of a solution that seamlessly integrates inflow control devices (ICDs) and permanent monitoring systems beneath a lateral access window marks a significant advancement in the field of intelligent completions.

By placing the PDHMS closer to the reservoir, the system enables the acquisition of high-resolution, real-time data from each lateral. This greatly enhances reservoir characterization and supports more accurate, data-driven production planning. The innovative configuration also delivers substantial operational efficiencies—reducing rig time by approximately 70% in this case study when compared to conventional methods that require multiple trips to install both the lateral access window and the completion hardware.

The ability to install the entire system in a single trip enhances wellbore integrity by minimizing connection points that could serve as potential leak paths. Additionally, the design eliminates the need for a permanently installed isolation sleeve, simplifying future interventions and improving lateral accessibility.

Production is further optimized by allowing direct flow through the window, reducing the complexity of future re-entry operations.

Conclusion

The system was deployed with exceptional service quality, achieving zero Non-Productive Time (NPT) and zero Cost of Poor Quality (COPQ). This flawless execution demonstrated the reliability of the integrated system and set a new benchmark for intelligent multilateral completions in complex reservoir environments. It not only underscores the system's reliability but also position it as a compelling benchmark for future deployments. Its inherent versatility and proven performance make it an ideal solution for complex well architectures, particularly in environments where multilateral access, real-time permanent monitoring, and precision flow control are critical to long-term field success. With this achievement, the foundation is firmly laid for widespread adoption across diverse assets in the Middle East and beyond.

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